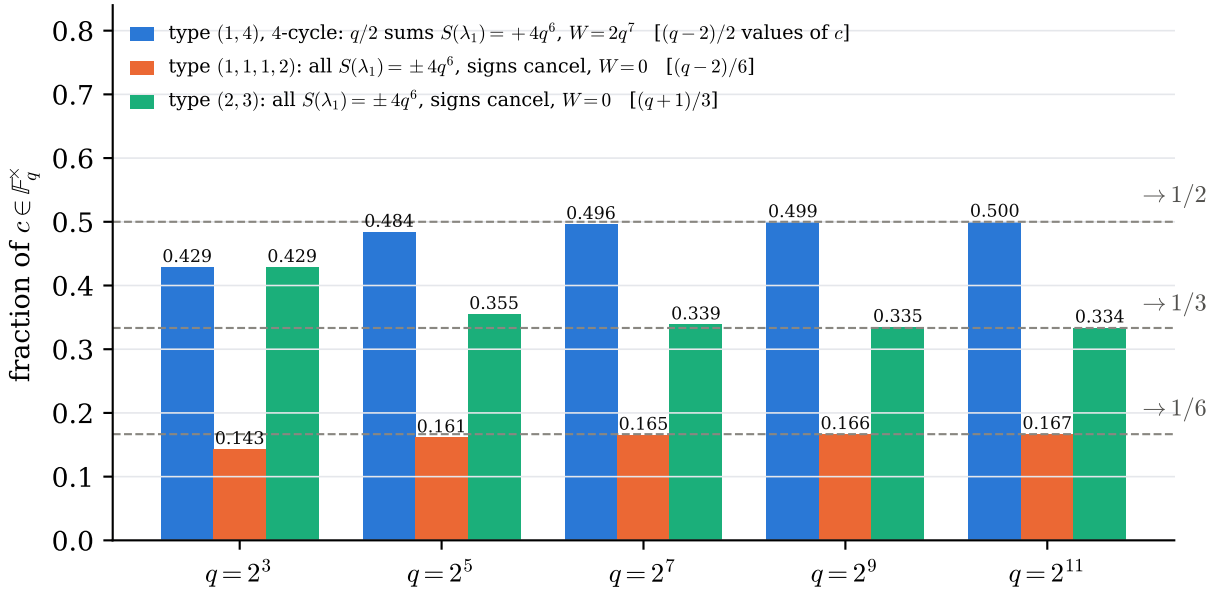
Factorisation type of v^5+v+c over $\mathbb{F}_q$, k odd, and the mixed term W
type $(1,4)$, 4-cycle: $q/2$ sums $S(\lambda_1)=+4q^6$, $W=2q^7$ [$(q-2)/2$ values of c]
type $(1,1,1,2)$: all $S(\lambda_1)=\pm 4q^6$, signs cancel, $W=0$ [$(q-2)/6$]
type $(2,3)$: all $S(\lambda_1)=\pm 4q^6$, signs cancel, $W=0$ [$(q+1)/3$]
fraction of $c\in\mathbb{F}_q^\times$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
$\to 1/2$
$\to 1/3$
$\to 1/6$
0.429
0.143
0.429
0.484
0.161
0.355
0.496
0.165
0.339
0.499
0.166
0.335
0.500
0.167
0.334
$q=2^3$
$q=2^5$
$q=2^7$
$q=2^9$
$q=2^{11}$